MG-ACC-46 Huawei Access Node Routing Protocol Flap Resolution SOP

Node: MG-ACC-46 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_9845

Description

The MG-ACC-46 Huawei Access Node at the MG Road Core site is experiencing a critical failure in the routing protocol domain, specifically a BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This failure domain occurs due to the complex interaction between the Huawei Access Node's Routing Information Base (RIB) and Forwarding Information Base (FIB), which are responsible for maintaining the routing table and forwarding packets, respectively. When the BGP route churn occurs, it can lead to a temporary loss of OSPF adjacency, causing the ISIS LSP to flood and ultimately resulting in the routing protocol flap. This situation can be exacerbated by hardware or software issues, such as faulty components or misconfigured protocols, which can further destabilize the routing protocol domain. The Huawei Access Node's advanced routing capabilities, including support for multiple routing protocols and advanced BGP features, can also contribute to the complexity of the issue. To resolve this failure, it is essential to understand the underlying mechanisms and interactions between the various routing protocols and hardware components.

Pre-requisite Checks

- Verify that the MG Road Core site has been notified and that all necessary personnel are aware of the situation.

- Check the status of the redundant links and ensure that they are functioning correctly.

- Ensure that the Huawei Access Node is powered on and that all necessary interfaces are operational.

Potential Root Causes

Cause	Probability
Faulty BGP configuration or misconfigured peers	High
Hardware issues with the Huawei Access Node or its interfaces	Medium
Software issues with the Huawei Access Node's routing protocols or firmware	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp process, Expected Output: BGP process is running with multiple peers connected., Description: This command displays the BGP process and its peers, which can help identify any misconfigured peers or faulty BGP configuration.

Command: display ospf neighbor, Expected Output: Multiple OSPF neighbors are connected and operational., Description: This command displays the OSPF neighbors and their status, which can help identify any OSPF adjacency loss issues.
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Command: display isis lsp, Expected Output: ISIS LSP flooding is occurring due to BGP route churn., Description: This command displays the ISIS LSP and its status, which can help identify any ISIS LSP flooding issues.
```

Resolution Steps

Step 1: Step: Verify BGP configuration and peers, Command: display bgp peer, Description: Check the BGP peers and configuration to ensure that they are correct and not causing the issue.

Step 2: Step: Reset BGP process, Command: reset bgp process, Description: Reset the BGP process to clear any temporary issues or misconfigurations.

Step 3: Step: Check OSPF adjacency and ISIS LSP, Command: display ospf neighbor and display isis lsp, Description: Verify the OSPF adjacency and ISIS LSP to ensure that they are stable and not causing the issue.

Post-Check Verification

```
Command: display bgp process, Expected Output: BGP process is running with multiple peers connected., Description: Verify that the BGP process is running correctly and that all peers are connected.

Command: display ospf neighbor, Expected Output: Multiple OSPF neighbors are connected and operational., Description: Verify that the OSPF neighbors are connected and operational.

Command: display isis lsp, Expected Output: ISIS LSP is stable and not flooding., Description: Verify that the ISIS LSP is stable and not flooding.
```

Roll-back Procedures

- Step: Restore previous BGP configuration, Command: restore bgp configuration, Description: Restore the previous BGP configuration to its last known good state.

- Step: Reset OSPF process, Command: reset ospf process, Description: Reset the OSPF process to its last known good state.

- Step: Reset ISIS LSP, Command: reset isis lsp, Description: Reset the ISIS LSP to its last known good state.